Tanzim Hossain Romel

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Uttara, Dhaka, Bangladesh

Software engineer and researcher with 3+ years of backend and AI engineering experience, building reliable AI-agent, developer-tooling, and healthcare analytics systems.

Research Interests

AI for Software Engineering (AI4SE), Software Engineering for AI (SE4AI), Trustworthy AI, LLM Agent Security, Empirical Software Engineering, Developer Tooling

Education

- **University of Alberta** — *Sep 2026*
  Incoming M.Sc. in Computing Science — Edmonton, Canada
  - Incoming thesis-based M.Sc. student in the U-A-Goose research group, supervised by Dr. Zhou Yang, with affiliation to Amii (Alberta Machine Intelligence Institute).
- **Bangladesh University of Engineering and Technology (BUET)** — *Mar 2018 - May 2023*
  B.Sc. in Computer Science and Engineering — Dhaka, Bangladesh
  - Selected coursework: Machine Learning, Operating Systems, Computer Security, Fault-Tolerant Systems, and High-Performance Database Systems.

Research Experience

- **University of Illinois Urbana-Champaign (UIUC)** — *Jun 2026 - Aug 2026*
  Remote Research Intern through UIUC++ SRSE 2026; Professor Tianyin Xu's lab
  - Working on software and systems reliability research with emphasis on rigorous empirical evaluation and reproducible research artifacts.
  - Preparing research prototypes, benchmark artifacts, and evaluation evidence for agentic software engineering and reliability tasks.
  - Building SRE-style benchmark artifacts around fault injection, mitigation oracles, shortcut rejection, and lifecycle evidence for Kubernetes reliability scenarios.
- **An Empirical Study on Remote Code Execution in Machine Learning Model Hosting Ecosystems** — *Jun 2025 - Oct 2025*
  Tools: Python, Bandit, CodeQL, Semgrep, YARA | Under review at ICSE 2027; preprint: arXiv:2601.14163
  - Conducted a large-scale study of ~45,000 ML model repositories across Hugging Face, ModelScope, OpenCSG, OpenMMLab, and PyTorch Hub.
  - Detected unsafe deserialization, eval-injection, malware-signature, and platform-safety risks using Bandit, CodeQL, Semgrep, YARA, and CWE-based classification.
  - Analyzed 600+ developer discussions to build a taxonomy of misconceptions around remote code execution, trust flags, and model-loading safety.
  - Compared platform defenses such as warning systems, scanning, SafeTensors support, and `trust_remote_code` controls to connect repository-level findings with ecosystem-level mitigation gaps.
- **The Choice Can Be the Attack: Auditing Aligned Backdoors in LLM Agents** — *Aug 2025 - Present*
  Tools: Python, vLLM, PyTorch, pandas, NumPy | Submitted to TACL 2026
  - Studied a hidden backdoor risk where a trigger changes which valid option an LLM agent chooses while the final answer still appears correct.
  - Built **SHIFT**, which reruns the same task with and without a known trigger, then checks whether the changed choice favors the attacker's target rather than normal reasons like price or rating.
  - Instrumented structured choice tasks with valid-option sets, logged option features, and matched no-trigger/trigger endpoint runs so the audit does not require model weights, logits, or hidden states.
- **Multi-Agent Framework for Generating Relational DB Schema & ERD** — *Jul 2025 - Present*
  Tools: Python, LangGraph, StateGraph, Z3 Solver, SQLAlchemy, Text2Schema
  - Extended SchemaAgent with Dr. Sukarna Barua (BUET) to turn natural-language requirements into relational schemas and ER diagrams.
  - Built a LangGraph pipeline for entity extraction, relationship detection, normalization checks, Z3 validation, and targeted retry.
  - Designed verification gates for entity/relation completeness, functional dependencies, normal-form compliance, key soundness, requirement coverage, and executable DDL repair.

Selected Research Software and Systems Projects

- **ctxhelm + HelmBench: Context Compiler and Source-Free Agent Evaluation** *May 2026 - Present*
  Tools: Rust, MCP, CLI, Git, Hybrid Retrieval, Symbol Indexing, JSON Schema, Agent Tracing
  - Built a local-first, read-only context compiler for coding agents, combining lexical search, symbol search, dependency expansion, related-test mapping, git co-change hints, semantic metadata, and memory signals.
  - Shipped ctxhelm releases with archive/Homebrew installation, checksum verification, doctor checks, MCP smoke tests, and source-free evaluation artifacts.
  - Built HelmBench as active evaluation infrastructure with trace schemas, privacy checks, evidence bundles, matrix runs, and quality gates that separate context-selection quality from downstream patch quality without exposing source, prompts, transcripts, or terminal logs.
- **Yet Another C Compiler** *Jun 2021 - Aug 2021; revised Oct 2024*
  Tools: C++17, Flex, Bison, CMake, x86-64 Assembly, SSA/IR Optimization
  - Built a C compiler with lexer/parser, semantic analysis, x86-64 code generation, and a linear-scan register allocator.
  - Modernized the Flex/Bison prototype with modular IR passes and SSA optimizations including SCCP, GVN, LICM, and dead-code elimination.
- **Bangla Handwritten Digit Recognition** *Jan 2023 - Feb 2023*
  Tools: Python, NumPy, OpenCV, CNN from Scratch, NumtaDB
  - Implemented a NumPy CNN from scratch to recognize handwritten Bangla digits from the NumtaDB dataset.
  - Built the preprocessing and training pipeline, reaching **95.9% accuracy** and placing **2nd among 120 students**.

Open Source Contributions

- **Selected Upstream Contributions** *2025 - 2026*
  Selected upstream contributions across SRE benchmarks, program analysis, .NET, web servers, agent systems, and compilers GitHub
  - **SREGym**: authored and hardened Kubernetes SRE benchmark work, including scheduler priority-preemption and Calico route-reflector label-drift scenarios with fault/oracle design, validation checks, and reviewer-facing evidence.
  - **RefactoringMiner**: merged local MCP tooling, WebDiff workflow support, merge-parent-aware commit diffs, assertion migration matching, and single-page PR viewed toggles for refactoring-analysis workflows.
  - **EF Core**: merged targeted data-stack changes for runtime migration creation/application, `ON DELETE SET DEFAULT` support, and nullable complex-property reload handling.
  - **GenHTTP**: added automatic request-body decompression and binary-response support with focused framework behavior tests.
  - **deepagents**: merged a CLI fix that stops and clears `LoadingWidget` animation timers on stop and unmount, preventing interval leaks in long-running terminal sessions and adding regression coverage for both cleanup paths.
  - **TypeScript**: merged an upstream-aligned declaration-emit alias-resolution fix that preserves imported type aliases for inferred exports across module boundaries in Microsoft's Go port of TypeScript.

Professional Experience

- **IQVIA** *Jun 2023 - Jun 2026*
  Software Development Engineer Dhaka, Bangladesh
  - Built and maintained KPI Library microservices in C#/.NET and EF Core on AWS for healthcare analytics workflows over patient-level datasets, including API, filtering, aggregation, and export paths used by global pharmaceutical clients.
  - Refactored a complex healthcare-analytics filter path into reusable query components, reducing duplicated business rules and improving filtering performance by 70% in the targeted workflow.
  - Reworked selected high-latency analytics paths with PostgreSQL query planning, targeted indexing, materialized views, Redis caching, and MongoDB tuning, reducing complex query time by 60% and API latency by 40%.
  - Designed Playwright-backed export regression tests with multi-layer hashing to compare generated artifacts deterministically, raising export-workflow coverage from 72% to 95% and catching content-level regressions.
  - Added OpenTelemetry and Jaeger tracing around API, query, and cache paths, making latency and regression sources visible across distributed services.
  - Built LangGraph/LangChain workflows with MCP, RAG, and `deepagents` to map analyst prompts into structured dashboard setup, modification, drill-down analysis, and validated export actions, reducing setup effort by 85% in the targeted workflow.
  - Optimized CI/CD build stages and parallel test execution, reducing deployment time by 35%; led POC work, mentored junior developers, and received the IQVIA Impact Program Silver Award for essential feature delivery.

Skills

Primary: Python, C#/.NET, Rust, TypeScript, Go, PostgreSQL, AWS, Docker, Kubernetes, LangGraph, MCP
Research and agent tooling: CodeQL, Semgrep, Bandit, YARA, Z3, RefactoringMiner, deepagents, vLLM, JSON Schema, source-free tracing
Systems and ML tooling: C++, Java, Flex, Bison, CMake, NS-3, NumPy, OpenCV, PyTorch, Linux namespaces
Backend and production tooling: EF Core, SQLAlchemy, REST APIs, Redis, MongoDB, SQL Server, DynamoDB, OpenTelemetry, Jaeger, Playwright

Honors and Awards

- **2nd Place – Bangla Handwritten Digits Recognition**: 95.9% accuracy, BUET ML Lab contest *2022*
- **Dean's List Award**: academic performance, BUET Level 2
- **National Science Olympiads**: prize winner in Bangladesh Physics and Chemistry Olympiads *2017*
- **Talentpool HSC Scholarship**: 15th in Rajshahi Board with 95.6% marks *2017*

References

References available upon request.